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RICHARD BIJU MELATH

Passport: AE397911 | **Date of birth:** 03/09/2003 | **Place of birth:** Angamaly, India |

Nationality: Indian | **Gender:** Male | **Phone number:** (+91) 9544211720 (Mobile) |

Email address: richardbijumelath48@gmail.com | **Website:** https://richardbmelath.vercel.app/ |

LinkedIn: www.linkedin.com/in/richardbmelath | **GitHub:** https://github.com/richard-b-melath |

Address: Ernakulam, Kerala, India (Home)

ABOUT ME

Computer Science graduate specializing in Artificial Intelligence with practical experience in developing AI-driven applications and scalable backend systems. Experienced in Python, FastAPI, React, MongoDB, Docker, machine learning, and large language models through software development, internships, and academic projects. Co-author of research publications in artificial intelligence and deep learning, with experience developing AI solutions for healthcare, waste management, and conversational applications. Interested in pursuing advanced studies in Artificial Intelligence, Data Science, and intelligent systems.

WORK EXPERIENCE

SOFTWARE DEVELOPER – DOCTORASSIST.AI (SARASWATHA HEALTHCARE RESEARCH PVT. LTD.) – 29/10/2025 – 27/02/2026 – BANGALORE, INDIA

- Developed features for an AI-driven medical software platform in a startup environment.
- Designed and implemented backend services with FastAPI and frontend components with React.
- Worked with MongoDB for database management and application data handling.
- Containerized services using Docker to support scalable deployment.
- Contributed to the design of a new scalable system architecture, laying the foundation for its implementation.

INTERNSHIP – SENSCRIPT TECHNOLOGIES PVT LTD – 02/06/2025 – 23/10/2025 – KOCHI, INDIA

- Developed a system to extract structured and unstructured data from PDF documents using Python libraries such as PDFPlumber and Tabula.
- Worked on processing and organizing extracted data for use in downstream applications and workflows.
- Built an automated email processing system that analyzes incoming emails and generates responses.
- Gained practical experience working with Python-based automation and AI-related tools in a real project environment.
- Collaborated with the development team to test and improve the reliability of the automation system.

INTERNSHIP – INFORMATION KERALA MISSION – 07/08/2024 – 06/01/2025 – INDIA

- Developed an AI-powered chatbot for the Harithamithram waste management application to assist users with service-related queries.
- Integrated ChatGPT API with structured data sources to generate accurate and context-aware responses.
- Implemented features for handling FAQs, providing service information, and supporting issue reporting within the application.
- Worked on improving user interaction and accessibility of waste management services through conversational AI.
- Collaborated with the development team to test and refine chatbot responses and functionality.

EDUCATION AND TRAINING

01/10/2026 – CURRENT Germany

M.SC. DATA SCIENCE Universität Trier

02/03/2026 – 28/08/2026 Kurumassery, India

GERMAN LANGUAGE (CURRENTLY IN PREPARATION (B1)) Little Flower Academy

Website https://littlefloweracademy.in/

01/08/2021 – 31/05/2025 KALADY, India

BACHELOR OF TECHNOLOGY (B.TECH) IN COMPUTER SCIENCE AND ENGINEERING – ARTIFICIAL INTELLIGENCE Adi Shankara Institute of Engineering and Technology (affiliated to APJAKTU)

Artificial Intelligence, Machine Learning, Data Structures and Algorithms, Database Management Systems, Software Engineering, Computer Networks, Operating Systems, Web Development, Python Programming, Deep Learning, and AI-based application development.
Final GPA: 8.52

Level in EQF EQF level 6

01/06/2019 – 31/03/2021 MAMBRA, India

HIGHER SECONDARY CERTIFICATE (SCIENCE STREAM) Union Higher Secondary School (UHSS) Mambra

Physics, Chemistry, Mathematics, Biology (Botany and Zoology) and English. Developed strong analytical thinking, problem-solving skills, and a solid foundation in scientific studies.
Final Score: 97.58%

Level in EQF EQF level 4

01/06/2018 – 31/03/2019 MAMBRA, India

SECONDARY SCHOOL LEAVING CERTIFICATE (SSLC) Little Flower Convent Higher Secondary School (LFCHSS) Koratty

General secondary education including Mathematics, Science, Social Science, and Languages.
Final Grade: Full A+ (Highest grade in all subjects)

Level in EQF EQF level 3

18/09/2024 – 02/10/2024 KALADY, India

CERTIFICATE COURSE ON ARTIFICIAL INTELLIGENCE Infosys Foundation Finishing School for Employability Program & ICT Academy

Fundamentals of Artificial Intelligence, machine learning concepts, and practical AI development using Microsoft Azure AI platform. Hands-on experience working with AI services and tools for building intelligent applications.

PROJECTS

Shankarani – AI Chatbot for College Digital Experience

Developed “Shankarani”, an AI-powered chatbot to enhance the digital experience for students and stakeholders at Adi Shankara Institute of Engineering and Technology.

- Built an intelligent chatbot to handle queries related to academic and institutional information
- Designed conversational workflows to improve user interaction and accessibility of information
- Collaborated with the college’s web team to support integration with existing digital systems
- Received a Certificate of Appreciation from the institution for the contribution
- The success of this project inspired the formation of DotGen, an early-stage AI venture focused on building practical AI solutions.

AI Courtroom Simulation Platform (LegalTech Multi-Agent System)

Developed an AI-powered multi-agent platform at the Codex Community Hackathon (Kochi) to make legal understanding more accessible and interactive.

- Designed a system to analyze legal case documents and extract structured elements such as facts, parties, claims, and evidence
- Built a multi-agent simulation with roles like judge, lawyers, and witnesses to mimic courtroom proceedings
- Enabled generation of arguments, objections, decisions, and summaries for better case comprehension
- Created an interactive “AI courtroom sandbox” to simplify legal reasoning and case flow visualization

Waste Management Personal Assistant

Technologies: Python, LLMs, Gemini, ChatGPT, Fine-tuning
Designed and developed an AI-powered waste management chatbot for the Harithamithram platform of the Kerala Government to assist users with service-related queries and information. The system provides accurate waste collection schedules, supports issue reporting, and offers guidance on sustainable waste management practices. Built using large language models to improve response accuracy and enhance user interaction with municipal waste management services.

Project Sustainability Analyser

Technologies: GenAI, Python, HTML, Flask, IBM Watsonx, Prompt Engineering
Developed a website using Flask and IBM Watson tools to predict sustainability issues in project ideas. The platform also recommends sustainable alternative products, promoting environmentally conscious decision-making

SKILLS

Technical Skills

Python (computer programming) | Machine Learning | Generative AI | Deep Learning | Conversational AI | Large Language Models (LLMs) | AI Application Development | API Python frameworks: Flask, FastAPI - Uvicorn | Stack Frontend (HTML, CSS, Javascript) | Javascript (React.js) | Databases (MongoDB MySql PostgresSql) | Dockerization with Docker (Intermediate) | Git/Github, Docker, Gitlab | Version Control (Git) | GOOGLE CLOUD (INTERMEDIATE) | System Design (Intermediate)

Soft Skills

Problem Solving | Design Thinking | Team Work | Leadership | Creative Thinking

LANGUAGE SKILLS

Mother tongue(s): **MALAYALAM**

Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	C1	C1	B2	B2	B2
GERMAN	A2	A2	A2	A2	A2

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

PUBLICATIONS

2024
Automating Tomorrow: The Convergence of AI and Robotics

Gayathri Dili, Sarika S., Akshay S., Richard B Melath, Amrutha Muralidharan Nair, Sharika T. R. (2024). International Journal of Intelligent Systems and Applications in Engineering (IJISAE)

Authors: Gayathri Dili, Sarika S., Akshay S., Richard B Melath, Amrutha Muralidharan Nair, Sharika T. R. | **Journal Name**: International Journal of Intelligent Systems and Applications in Engineering (IJISAE) | **Publisher**: International Journal of Intelligent Systems and Applications in Engineering

Link https://ijisae.org/index.php/IJISAE/article/view/6543/5392

2025
Harnessing Deep Learning for Early Detection of Plant Diseases: A Comprehensive Approach

Amrutha Muralidharan Nair, Devika P Sajith, Pavithra Deepu E, Madhavan V, Sarang R, Thomas Eldhose, Richard B Melath . Harnessing Deep Learning for Early Detection of Plant Diseases: A Comprehensive Approach. IEEE Conference Proceedings.

Authors: Dr. Amrutha Muralidharan Nair, Devika P Sajith, Pavithra Deepu E, Madhavan V, Sarang R, Thomas Eldhose, Richard B Melath | **Journal Name**: IEEE Conference Proceedings | **Publisher**: IEEE

Link https://ieeexplore.ieee.org/abstract/document/11168593

VOLUNTEERING

01/06/2016 – 31/03/2019 KORATTY
STUDENT POLICE CADET

01/06/2019 – 31/03/2021 MAMBRA
NATIONAL SERVICE SCHEME (NSS)

CERTIFICATIONS

Infosys Foundation and ICT Academy
Certificate Course on Artificial Intelligence

GPT3 Cloud Innovations
Generative AI-30 hours Add-on Course

NPTEL
Problem solving through programming in C

NPTEL
Fundamentals of Artificial Intelligence

NPTEL
Computer vision and image processing - fundamentals

MathWorks
Matlab Onramp Certification

Infosys Springboard
Computer Vision 101

Infosys Springboard
Artificial Intelligence Foundation

ACHIEVEMENTS

Developed Shankarini, an AI-powered college chatbot that dynamically extracts website data to provide accurate and professional responses to student queries.

Designed and developed the official website brahma25.live for Brahma25, the national-level inter-college technical and cultural fest organized by ASIET.

Our project shortlisted and participated in IEDC innovation challenge

Shortlisted and participated in IBM's watsonx ai hackathon conducted at IBM kochi office.

Developed website for AI department fest Prayag24